

CLOWN

Bubble-shooting sleeve — build manual

Squeeze your fingers into your palm and a comet of light runs from your elbow to your fingertips, arriving exactly as bubbles start firing out of your hand. Two arms, two identical rigs.

This page is every part and what it actually does. Each one is numbered. The assembly steps from page 2 onwards refer back to those numbers in **[brackets]**. All quantities are **PER ARM** — you are building two of everything.

THE BUBBLES

1. Bambu Lab Electric Bubble Maker Kit 01 (P6M) — x1

A tiny fan that blows air across a film of soap. You are not modifying it — you are just taking over the decision of *when* it switches on. Each kit also includes a battery [8] and a charger.

THE LIGHT

2. WS2812B LED strip, 60 LED/m, 5V, black PCB, IP65 — 0.4 m

A ribbon where every single light has its own chip inside, so each one takes orders individually down one data wire. That is the only reason a travelling comet is possible — an ordinary strip can only be one colour at a time. IP65 is the clear silicone sleeve that keeps the soap out.

3. Thin, pale, stretchy glove — x1

The strip hides under this. Fabric spreads the light into a smooth glow instead of visible dots — but only if it is thin and pale. Thick or dark fabric, especially leather, swallows almost all of it.

THE BRAIN

4. Seeed XIAO ESP32-C3 — x1

A complete computer the size of a postage stamp. It watches the trigger and decides what all 21 lights do, about 60 times a second. You reprogram it with a USB-C cable.

5. 74AHCT125 level shifter — x1

A translator that also shouts. The brain speaks at 3.3V; the strip is listening for 5V. Usually the strip *sort of* hears it anyway — which is exactly the problem. Leave this out and your lights flicker randomly, normally for the first time once you are in costume.

MOTOR CONTROL

6. N-channel logic-level MOSFET module — x1

An electric switch with no moving parts. The brain can think, but it cannot push enough power to spin a motor, so it flicks this instead and this lets the big current through.

7. 1N5819 flyback diode — x1

A motor is really just a coil of wire, and cutting power to a coil makes it kick a nasty voltage spike backwards. This is a one-way valve that gives the spike a safe loop to burn itself out in. It costs 20 cents; skip it and the MOSFET [6] eventually dies.

POWER

8. Protected 18650 cell — x1 + spares

A rechargeable battery — one comes free with each bubble kit [1]. *Protected* means a guardian circuit inside cuts it off before you over-drain or over-charge it. Do not substitute cheaper unprotected cells. This is strapped to your arm.

9. 18650 sled with wire leads — x1

The slot the battery sits in, so you can swap a flat cell for a fresh one without a soldering iron. Print one with a lid that clicks shut.

10. 5V boost converter module — x1

A pump, but for voltage. The battery only makes about 3.7V and the lights and brain need 5V. Here it only has to carry the lights (~0.3A) because the motor runs straight off the battery, so a small cheap one is genuinely fine.

11. 2A resettable polyfuse (PPTC) — x1

A safety valve that resets itself. If something shorts out, it suddenly becomes very resistant and chokes the current off before anything gets hot. Let it cool and it goes back to normal by itself.

12. 100 kilo-ohm resistors — x2

Two resistors that shrink the battery voltage neatly in half, so the brain [4] can measure it without being damaged. This is what lets the costume warn you that the cell is nearly flat instead of just dying.

SIGNAL CONDITIONING

13. 1000 uF electrolytic capacitor — x1

A tiny water tank for electricity. When a lot of LEDs switch on in the same instant they all gulp power at once; the tank smooths out the gulp so the voltage does not dip. Polarised — the leg with the stripe is negative, and backwards it pops.

14. 330-470 ohm resistor — x1

A speed bump on the data wire. It softens the sharp edge of the signal so it does not bounce back down the wire and garble the message to the first pixel. Skip it and pixel 1 misbehaves while all the others are fine.

THE TRIGGER

15. Snap-action lever microswitch — x1

A clicky button with a little metal arm sticking off it. The arm is the whole point: it gives you a wide target, so you can fire mid-performance without looking or aiming. It also clicks, so you feel it go.

WIRE AND MATERIALS

16. Silicone hookup wire, 22-26 AWG — as needed

Wire with soft rubbery insulation instead of stiff plastic. It survives thousands of bends. Ordinary wire — especially solid-core — snaps at the elbow and wrist within a few hours of wearing it. Thicker for power, thinner for signals.

17. Heat-shrink tubing, assorted — as needed

Plastic sleeve that shrinks tight when heated, sealing and insulating each solder joint. Slide it onto the wire **BEFORE** you solder. Everyone forgets once.

TOOLS

18. Multimeter — x1

For checking polarity before you connect the battery the first time. Two minutes here saves re-ordering a fried brain [4] and strip [2].

19. Soldering iron, solder, flux — as needed

For all of the above. Work somewhere ventilated; flux fumes are unpleasant.

20. 3D printer — x1

For the trigger plate, the battery sled [9] and the controller pod. You already have one.

Before you order

Buy two bubble kits [1]. You need one blower per hand, and each kit also brings a protected cell [8] and a charger — that covers both arms without buying batteries separately.

Test your glove fabric [3] first. Hold a lit strip [2] under it in a dark room. If the glow is weak, change gloves — no setting in the firmware rescues fabric that eats the light.

Assembly

Read this whole manual once before picking up the soldering iron [19]. Several steps are much easier if you know what is coming — notably that heat-shrink [17] goes on *before* you solder, and that you bench-test everything *before* anything goes inside a glove [3].

Budget a full afternoon for the first arm. The second takes about half as long, because by then you know what you are doing.

Safety, briefly but seriously

The cell [8] is lithium. Never short its terminals, never charge a puffy or damaged cell, and do not leave it charging unattended.

Check polarity with the multimeter [18] before you connect the battery the first time — reversed power destroys the brain [4] and the strip [2] instantly and permanently.

Step 1 — Measure your arm

Put on the glove [3] and hold your arm out in the finger-gun pose. Measure and write down:

- **Elbow to wrist crease** — typically about 25 cm.
- **Wrist crease to knuckles**, across the back of the hand — about 10 cm.
- **Where your middle and ring finger pads land** on your palm when you squeeze. That is where the trigger [15] goes. Mark it with a fabric pen.

Convert the first two into pixel counts. At 60 LED/m each pixel on the strip [2] is **1.67 cm**:

```
forearm pixels = forearm cm / 1.67
hand pixels    = hand cm / 1.67
```

That is usually about **15** forearm pixels and **6** hand pixels. Round down — slightly short is fine, too long does not fit. Keep these numbers; you type them into the firmware in Step 8.

Step 2 — Cut the strip

The strip [2] has marked cut lines between every pixel, with copper pads either side. **Cut through the middle of the pads**, not beside them — you need pad left on both pieces to solder to. Cut two pieces to the pixel counts from Step 1.

Check the arrows before you cut

The strip has small printed arrows showing which way data flows. Data must run **elbow to fingertips**, so both pieces need their arrows pointing the same way, toward your fingers. Getting this backwards is the most common build mistake and the fix is unsoldering everything. Mark the elbow end with tape — that end is pixel 0.

Why two pieces? Your wrist bends constantly, and a single continuous strip across that joint cracks its internal copper traces within hours of wear. Everything past the crack then goes dark. The flexible jumper in Step 3 absorbs that movement instead.

Step 3 — Solder the wrist jumper

Join the two strip pieces with about 4 cm of silicone wire [16], three conductors: 5V, GND and data.

- Slide heat-shrink [17] onto each wire **now**, before soldering.
- Peel back a little silicone sleeve at each strip end to expose the pads.
- Tin the pads and the wire ends separately, then join them.
- Forearm piece **output** end to hand piece **input** end: 5V to 5V, GND to GND, **DO to DI**.
- Shrink the tubing over each joint, then seal the exposed strip ends with hot glue or clear silicone. That is the soap-proofing.

Make the jumper slightly longer than the gap needs, so there is a small service loop — it must never be pulled tight when you fully extend your wrist. Tug each wire gently to test the joints now, while they are still reachable.

Step 4 — Build the controller pod

This all lives on a small piece of protoboard that sits on your **upper arm**, above the soap spray.

From	To	Notes
Battery [8] +	Polyfuse [11], then everything else	Fuse goes first, right at the cell
Battery [8] –	Common ground	Everything shares this

From	To	Notes
Battery + (after fuse)	Boost [10] IN+	
Boost [10] OUT+ (5V)	XIAO [4] 5V pad, 74AHCT125 [5] Vcc, strip [2] 5V	
Battery + (after fuse)	Motor +	Motor runs direct from the cell , not from 5V
Motor –	MOSFET [6] output	
XIAO D2 (GPIO4)	MOSFET [6] gate input	
XIAO D10 (GPIO10)	74AHCT125 [5] input	
74AHCT125 [5] output	Resistor [14], then strip DIN	Resistor close to the strip
XIAO D1 (GPIO3)	Microswitch [15], other side to ground	
XIAO D0 (GPIO2)	Midpoint of the two 100k resistors [12]	Battery monitor

The battery monitor divider

Battery + (after fuse) —[100k]—+—[100k]— Ground
 |
 XIAO D0

The midpoint sits at exactly half the cell voltage, which is safely inside what the brain [4] can measure.

Also fit

- The **1000 uF capacitor** [13] across the strip's 5V and ground, physically close to the strip. Watch polarity — the striped leg is negative.
- The **1N5819 diode** [7] directly across the motor's two terminals, banded end to **positive**. Backwards it is a dead short, so check this one twice.

Three rules that matter

1. Everything shares **one common ground** — brain [4], strip [2], level shifter [5], MOSFET [6], boost [10] and cell [8]. Skipping this causes bizarre, hard-to-diagnose behaviour.
2. **Do not use D8 or D9** (GPIO8/GPIO9) for anything. They are boot pins and the board will not start reliably.
3. **Leave the USB-C port reachable** when you design the printed pod. You will reflash this far more often than you expect — every timing tweak, every colour change. A cutout in the pod wall is enough.

Step 5 — Wire the trigger

- Solder thin silicone wire [16] to the microswitch's **COM** and **NO** (normally open) terminals.
- Run them up the forearm to the pod: one to D1, one to ground.
- Mount the switch [15] on a small printed plate [20], positioned so the **lever** sits under your middle and ring finger pads at the spot you marked in Step 1.

Orient the lever so a natural squeeze presses across its length. You should be able to fire it with your eyes closed, in one motion, every time. If you catch yourself aiming, reposition the plate until you do not.

Step 6 — Bench test, before anything goes in the glove

Do not skip this step

Sewing everything into a glove [3] and *then* discovering a reversed data arrow is genuinely miserable. Test it flat on the table first.

- **Check polarity with the multimeter** [18]. Battery + and – where you expect; boost [10] output close to 5.0V. Only then connect the brain [4].
- Power up. You should get the dim breathing idle glow.
- Press the microswitch [15] by hand. The comet should launch from the **elbow** end and travel to the **fingertip** end, and the motor should spin.
- If the comet runs backwards, the strip pieces are reversed. **Fix it in the wiring, not in the code** — both arms run identical firmware and that is worth a resolder.

- Let it run five minutes, then touch the boost [10] and MOSFET [6]. Warm is fine; too hot to touch means power down and look for a short.

Step 7 — Fit the bubble maker

Mount the blower [1] so its nozzle points **past your fingertips**, in the direction the comet travels. The bottle of solution sits on the back of the hand or forearm — the kit's MakerWorld collection has mounts to start from. Wire the motor back to the MOSFET [6] using silicone wire [16], because this run flexes.

Use the one-way valve cap that came with the kit. It is why the bottle does not dump solution when you swing your arm around.

Step 8 — Flash the firmware

- Install the Arduino IDE, then add ESP32 board support (Boards Manager, search esp32, install the Espressif package).
- Install the **FastLED** library from the Library Manager.
- Select board XIA0_ESP32C3.
- Open `firmware/clown_arm/clown_arm.ino`.
- **Edit the layout constants** at the top to match your Step 1 measurements.
- Plug in USB-C and upload. If it fails, hold BOOT while plugging the cable in.

```
constexpr int FOREARM_PX = 15; // your forearm pixel count
constexpr int HAND_PX    = 6;  // your hand pixel count
constexpr int GAP_PX     = 2;  // jumper length / 1.67 cm
```

Flash the identical sketch to both arms. There is deliberately no left/right setting — as long as pixel 0 is at the elbow on each arm, the mirroring is physical only.

Step 9 — Tune the timing

This is the step that makes the whole effect work, and it can only be done on the finished arm. The goal: **the comet reaches your fingertips at the exact moment the first bubble appears**. When it is right, the light looks like it is *causing* the bubbles. Wrong by even 100 ms and it just looks like lag.

- Film yourself firing it in slow motion — any modern phone does 120 or 240 fps.
- Watch it back frame by frame. Does the comet arrive early or late?
- Adjust COMET_TRAVEL_MS: comet arrives **before** the bubbles, increase it; **after** the bubbles, decrease it.
- Reflash and film again. Expect three or four rounds.

The default is 250 ms. Your real number depends on how fast your particular blower [1] spins up, which is why it is guesswork until you measure it.

Step 10 — Mount into the glove

Hiding the strip [2] under fabric is an upgrade, not a compromise — the fabric diffuses the pixels into one smooth continuous streak instead of visible dots. But it only works if the fabric cooperates.

Test the fabric before you commit

Hold a powered section of strip [2] under the glove [3] in a dark room. Thin, pale, stretchy fabric glows beautifully; thick or dark fabric, especially leather, swallows almost all of it. If you cannot see it clearly, no firmware setting will rescue it — get a different glove.

- Run the strip across the **back of the hand**, over the knuckles. Never the palm — that is where the trigger [15] is and where you grip things.
- Stop short of the fingertips. The bubble nozzle [1] needs that space, and finger joints flex too much for strip to survive.
- Sew a **fabric channel** for the strip to sit in rather than gluing it down flat. The channel lets it slide as your hand flexes; glued-taut strip tears itself off, or tears the glove.
- Add **strain relief** where wires cross the elbow and wrist — a small loop of wire tacked down each side of the joint, so movement pulls on the tack rather than on a solder joint.
- MAX_BRIGHTNESS is preset high to punch through fabric. You can raise it further for thicker gloves, but current draw climbs with it and battery life drops.

Step 11 — Build the second arm

Repeat Steps 1 to 10. The only thing to watch: **pixel 0 goes at the elbow on this arm too.** It is tempting to mirror the wiring along with the physical build. Do not. Identical wiring means identical firmware means one thing to maintain.

Troubleshooting

Symptom	Likely cause
Nothing lights at all	Check common ground. Check boost [10] output is about 5V. Check strip DIN is on the <i>input</i> end — follow the arrows.
First pixel wrong colour, rest fine	Missing or wrong-value data resistor [14].
Random flickering, especially when moving	Missing level shifter [5], or a loose ground.
Comet runs fingertips to elbow	Strip [2] is reversed. Fix the wiring, not the code.
Comet appears to jump at the wrist	GAP_PX does not match your actual jumper length.
Colours wrong, red and green swapped	Change GRB to RGB in the addLeds line.
Motor whines audibly	PWM frequency dropped below 20 kHz — check MOTOR_PWM_HZ.
Motor does not spin	MOSFET [6] gate not on D2, or it is not a logic-level part.
Everything dies when the motor starts	Cell [8] sagging, or the polyfuse [11] tripping. Check the charge.
Works, then dies after a few minutes	Cell [8] protection circuit cutting out. Recharge or swap.
Elbow pixel pulsing red	Low-battery warning. Swap the cell [8].
Board will not accept uploads	Hold BOOT while plugging in USB.
Strip goes dark past the wrist	Cracked trace or failed jumper joint — the failure the jumper exists to prevent. Replace it.

Changing how it behaves

The whole point of the strip [2] is that every pixel takes orders individually. Everything you would normally want to change lives in one marked block at the top of the sketch — edit, reflash over USB-C, done.

Constant	Changes
THEME_HUE	The comet's colour, 0-255 around the colour wheel
COMET_TRAVEL_MS	Comet speed, and its sync with the bubbles
COMET_REPEAT_MS	How rapid-fire the stream is while held
COMET_FADE	Tail length — higher is shorter
MAX_BRIGHTNESS	Overall brightness, for your glove's fabric
IDLE_BRIGHTNESS	Resting glow. Set to 0 for fully dark when idle
MOTOR_RUN_DUTY	Bubble rate
VBAT_WARN_MV	When the low-cell warning starts

The XIAO [4] has Wi-Fi and Bluetooth sitting unused, already paid for. Natural next steps, in rough order of effort: more patterns with a long-press to cycle them; ESP-NOW so one arm's trigger fires both arms at once; a small web page served by the board so you can change colour and speed from your phone without a laptop.

Field kit

What to carry when you actually wear this:

- A spare charged cell [8] per arm.
- A pre-soldered spare wrist jumper.
- A spare microswitch [15].
- **Extra bubble solution.** This runs out long before the battery does.
- A small screwdriver and some electrical tape.